

SIMATS ENGINEERING

ASSESSMENT TOOL 2- COMPARATIVE ANALYSIS

Course Code:	CSA12	Course Title:	Computer Architecture
CO Assessed:	CO4	Assessment:	ASSESSMENT TOOL2- COMPARATIVE ANALYSIS
Weightage:	30%	Total Marks:	25
CO4 Statement:	<i>Implement hierarchical memory systems by differentiating cache levels (L1, L2, L3), virtual memory with paging, and advanced memory technologies (DRAM, SRAM, NAND Flash, MRAM, RRAM) based on latency, bandwidth, and throughput characteristics.</i>		
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ANNEXURE A COMPARATIVE ANALYSIS

- 1.Compare L1, L2, and L3 cache levels in terms of latency, bandwidth, and throughput, and analyze their impact on processor performance.
- 2.Differentiate SRAM and DRAM by evaluating their latency, bandwidth, and throughput characteristics in cache and main memory design.
- 3.Compare virtual memory with paging and cache memory mechanisms in terms of access latency and system throughput.
- 4.Analyze the differences between NAND Flash and DRAM with respect to latency, bandwidth, and suitability for hierarchical memory systems.
- 5.Compare MRAM and RRAM technologies by examining their performance in terms of access time, throughput, and energy efficiency.

Code of Conduct:

*I Deepa preya H / 192421047 certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the student:

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Scope of Items	20%	Selects highly relevant items with clear scope and justification	Selects appropriate items with minor gaps	Selection is limited or partially relevant	Items are unclear or not relevant
Comparison Depth	25%	Provides detailed and comprehensive comparison across multiple aspects	Provides adequate comparison with some depth	Limited comparison with few aspects	Very minimal or no comparison
Use of Evidence	20%	Supports comparison with accurate data, examples, or references	Uses some supporting evidence with minor gaps	Limited or weak supporting evidence	No supporting evidence provided
Critical Thinking	20%	Demonstrates strong critical thinking with clear interpretation and reasoning	Shows reasonable interpretation with minor gaps	Basic interpretation; lacks depth	No meaningful interpretation
Clarity of Presentation	15%	Well-organized, clear, and logically structured presentation	Generally clear with minor issues	Somewhat unclear or poorly structured	Disorganized and difficult to understand

1. Compare L1, L2, and L3 Cache Levels in Terms of Latency, Bandwidth, and Throughput, and Analyze Their Impact on Processor Performance

Cache memory is a small and very fast memory located close to or inside the CPU. Its main purpose is to reduce the time required by the processor to access frequently used instructions and data. Modern processors generally use multiple levels of cache, namely **L1, L2, and L3**. These levels form a hierarchy in which smaller memories are faster and larger memories are comparatively slower.

L1 Cache

L1 cache is the smallest and fastest cache memory. It is usually located inside each processor core and is divided into instruction cache and data cache. Because L1 is physically very close to the CPU execution units, it provides extremely low access latency.

L1 cache generally has the highest bandwidth and lowest latency among the three cache levels. It stores the instructions and data that are most frequently required by the processor. When the required data is found in L1, it is called an L1 cache hit, and the CPU can continue execution with very little delay.

However, L1 cache has limited capacity because increasing its size can increase its access time and hardware complexity.

L2 Cache

L2 cache is larger than L1 but slightly slower. Depending on the processor architecture, L2 may be dedicated to each core or shared between a smaller group of cores.

When the required data is not available in L1, the processor checks L2. Since L2 is larger, it has a better chance of containing data that was not found in L1. However, its latency is higher than L1.

L2 provides a good balance between capacity, latency, bandwidth, and cost. It significantly reduces the number of requests sent to slower L3 cache or main memory.

L3 Cache

L3 cache is generally the largest cache among L1, L2, and L3. It is commonly shared among multiple processor cores.

Because L3 is larger and located farther from the individual execution units than L1, its access latency is higher. Its bandwidth is also generally lower than that of L1 and L2. However, it plays an important role in reducing accesses to main memory.

For a multicore processor, L3 can allow different cores to access commonly required data without repeatedly accessing DRAM.

Impact on Processor Performance

The cache hierarchy has a major impact on processor performance. Consider the access sequence:

CPU → L1 → L2 → L3 → Main Memory

If the requested data is present in L1, the processor gets it very quickly. If it is absent from L1 but available in L2, the processor experiences a slightly longer delay. If it is absent from both L1 and L2 but present in L3, the delay increases further. If the data is not available in any cache, the CPU must access main memory, which is considerably slower.

Therefore, a high cache-hit rate improves processor performance by reducing average memory-access time.

Example

Suppose a program repeatedly uses the same variables inside a loop. These variables are likely to remain in cache. The CPU can repeatedly access them from L1 or L2 instead of accessing DRAM every time. This reduces waiting time and increases instruction execution throughput.

Conclusion

L1, L2, and L3 caches work together to provide a balance between **speed and capacity**. L1 offers the lowest latency and highest bandwidth, L2 provides greater capacity with slightly higher latency, and L3 provides even greater capacity and reduces expensive main-memory accesses. Thus, an efficient cache hierarchy can significantly improve processor performance, particularly for applications with good data and instruction locality.

2. Differentiate SRAM and DRAM by Evaluating Their Latency, Bandwidth, and Throughput Characteristics in Cache and Main Memory Design

SRAM and DRAM are two important types of semiconductor memory used in computer systems. Although both are volatile memories, they differ significantly in their internal structure, speed, cost, density, and applications. SRAM is primarily used for cache memory, whereas DRAM is commonly used as the main memory of a computer.

SRAM

SRAM stands for Static Random Access Memory. It stores each bit using a flip-flop-based circuit. SRAM does not require periodic refreshing as long as power is continuously supplied. Because SRAM uses a relatively large circuit for each stored bit, it occupies more physical area and is more expensive. However, this design allows SRAM to provide very fast access. SRAM has very low latency, high bandwidth, and high throughput, making it suitable for processor caches.

DRAM

DRAM stands for Dynamic Random Access Memory. It stores data using a capacitor and transistor arrangement. The capacitor gradually loses its charge, so DRAM requires periodic refreshing.

The simpler cell structure allows DRAM to achieve much higher memory density at a lower cost. However, refresh operations and the nature of its storage mechanism result in higher latency compared with SRAM.

DRAM therefore provides large memory capacity at reasonable cost and is suitable for main memory.

SRAM in Cache Design

Cache memory must respond to CPU requests extremely quickly. Even a small increase in memory access latency can affect processor performance when memory is accessed frequently.

Therefore, SRAM is preferred for L1, L2, and many L3 cache implementations because its low latency allows the processor to retrieve frequently used data rapidly.

For example, when a CPU searches for an instruction, it first checks L1 cache. If the instruction is present, SRAM can provide it much faster than DRAM.

DRAM in Main Memory Design

Main memory requires much greater capacity than cache. A typical computer may require gigabytes of RAM, while processor caches are much smaller.

Using SRAM for the entire main memory would be extremely expensive and would require a large amount of chip area. DRAM provides a much better balance between capacity and cost. Therefore, DRAM is used as the main memory that stores currently running programs and their data.

Latency, Bandwidth and Throughput

Latency refers to the delay before data access begins or completes. SRAM has lower latency than DRAM, making it more suitable for cache.

Bandwidth refers to the amount of data that can be transferred per unit time. Both SRAM and DRAM can provide high bandwidth, but SRAM is optimized for very fast processor-facing access.

Throughput represents how effectively the memory system can handle repeated requests. SRAM provides excellent throughput for small, frequent accesses, whereas DRAM provides high throughput for larger main-memory workloads.

Example

Consider a processor running a program that repeatedly accesses a variable:

```
sum = sum + value
```

If the required data is stored in SRAM cache, the processor can access it very quickly. If the required data has to be retrieved from DRAM, the processor experiences greater memory-access latency.

Conclusion

SRAM and DRAM are complementary rather than competing technologies. SRAM is faster, more expensive, and less dense, making it ideal for cache memory. DRAM is slower but cheaper and denser, making it ideal for main memory. The combination of SRAM cache and DRAM main memory creates an efficient hierarchical memory system.

3. Compare Virtual Memory with Paging and Cache Memory Mechanisms in Terms of Access Latency and System Throughput

Virtual memory and cache memory are both important memory-management mechanisms, but they serve different purposes. Cache memory is designed primarily to reduce processor access latency, whereas virtual memory allows programs to use an address space larger than the available physical RAM and provides efficient memory management.

Virtual Memory

Virtual memory is a memory-management technique that provides each process with a logical or virtual address space. The operating system maps virtual addresses to physical memory addresses.

A common implementation of virtual memory is paging. In paging, the virtual address space is divided into fixed-size pages, while physical memory is divided into corresponding page frames.

When a required page is present in physical memory, it can be accessed normally. If the page is not present, a page fault occurs. The operating system must then retrieve the required page from secondary storage and place it into RAM.

This process introduces significant latency.

Cache Memory

Cache memory is a high-speed memory system positioned between the CPU and main memory. It stores recently or frequently accessed instructions and data.

Cache memory works mainly on the principle of locality of reference:

- **Temporal locality:** recently used data is likely to be used again.
- **Spatial locality:** data near recently accessed data is likely to be used soon.

Access Latency

Cache memory has extremely low access latency when there is a cache hit. This is important because processors operate much faster than main memory.

Virtual memory has a different performance characteristic. A normal access to a page already present in RAM may be reasonably fast, but a page fault can cause a very large delay because the operating system must access secondary storage.

Therefore:

Cache hit → **Very low latency**

Cache miss → **Higher latency**

Page fault → **Very high latency**

Effect on System Throughput

Cache memory improves throughput by allowing the processor to spend more time executing instructions instead of waiting for memory.

Virtual memory can reduce throughput when page faults occur frequently. A system experiencing excessive page faults may spend considerable time moving pages between RAM and secondary storage rather than executing useful instructions. This condition is known as thrashing.

Example

Suppose a program repeatedly accesses an array.

If the required data is already in cache, the CPU can access it quickly, resulting in high performance.

Now suppose the program requires a page that is not currently in RAM. A page fault occurs. The operating system has to retrieve the page from secondary storage, which takes much longer. During this time, the processor may have to wait.

Relationship Between the Two

Cache and virtual memory work at different levels of the memory hierarchy:

CPU → Cache → Main Memory → Secondary Storage

Cache attempts to make memory access faster, while virtual memory allows the system to manage memory efficiently and provide a larger logical address space.

Conclusion

Virtual memory with paging and cache memory solve different problems. Cache memory focuses on low access latency and high processor throughput, whereas virtual memory focuses on memory capacity, address-space management, and efficient use of physical memory. Cache misses cause moderate performance penalties, while page faults can cause very large delays. Both mechanisms are essential for building an efficient modern computer system.

4. Analyze the Differences Between NAND Flash and DRAM with Respect to Latency, Bandwidth, and Suitability for Hierarchical Memory Systems

NAND Flash and DRAM are both semiconductor memory technologies, but they are designed for different purposes. DRAM is a volatile memory primarily used as main memory, while NAND Flash is non-volatile and is mainly used for persistent storage, such as SSDs, memory cards, and USB storage.

DRAM

DRAM stores data using electrical charge in memory cells. Since the charge gradually leaks away, DRAM needs periodic refreshing.

Its main advantage is its relatively low latency and high performance compared with NAND Flash. This makes it suitable for situations where the CPU requires frequent random access to data.

DRAM is therefore placed relatively close to the CPU in the memory hierarchy.

NAND Flash

NAND Flash is a non-volatile memory technology. It retains stored data even when power is removed.

It provides high storage density and is more economical for large-capacity storage. However, its access latency is significantly higher than DRAM, particularly for random access and write operations.

NAND Flash is therefore more appropriate for persistent storage rather than direct CPU main-memory use.

Latency

Latency is one of the most important differences between DRAM and NAND Flash.

DRAM provides relatively fast random access, which is essential for a processor's main-memory operations.

NAND Flash has considerably higher access latency. Modern SSDs improve performance through controllers, parallel NAND channels, caching, and sophisticated management, but NAND Flash remains slower for direct random memory access than DRAM.

Bandwidth

DRAM provides high bandwidth for continuous data transfers between the processor and main memory.

NAND Flash devices can also provide substantial sequential bandwidth, especially when multiple flash chips and channels operate in parallel. However, the characteristics differ significantly from DRAM, particularly for random accesses and writes.

Throughput

DRAM can support frequent random accesses and is therefore well suited for CPU workloads.

NAND Flash can achieve good throughput for storage operations, especially sequential reads and writes. SSD controllers use parallelism to improve overall storage throughput.

Suitability in Hierarchical Memory

A typical memory hierarchy can be represented as:

CPU → Cache → DRAM → NAND Flash

DRAM acts as the main memory, while NAND Flash is normally positioned at the secondary storage level.

The reason is based primarily on the trade-off between speed and capacity.

- Cache: extremely fast but small
- DRAM: fast and larger
- NAND Flash: slower but much larger and non-volatile

Example

When a computer starts a program, the program is stored persistently on an SSD using NAND Flash. When the program is executed, required data is loaded into DRAM. Frequently used data may then be copied into CPU cache.

Thus, the same program can move through different memory levels depending on the current requirement.

Advantages of NAND Flash

1. Non-volatile storage
2. High storage density
3. No refresh operation
4. Suitable for large-capacity storage
5. Lower physical size compared with many traditional storage technologies

Advantages of DRAM

1. Low access latency
2. High memory bandwidth
3. Suitable for random access
4. High write endurance
5. Appropriate for active program execution

Conclusion

DRAM and NAND Flash occupy different positions in the memory hierarchy because they provide different performance characteristics. DRAM offers lower latency and high throughput, making it suitable for main memory. NAND Flash provides high capacity and non-volatility, making it suitable for secondary storage. A hierarchical memory system combines both technologies to achieve a balance between performance, capacity, cost, and persistence.

5. Compare MRAM and RRAM Technologies by Examining Their Performance in Terms of Access Time, Throughput, and Energy Efficiency

MRAM and RRAM are emerging non-volatile memory technologies that have attracted attention as alternatives or complements to conventional memory technologies. Both can retain information without continuous power, but they use different physical mechanisms for storing information.

MRAM

MRAM stands for Magnetoresistive Random Access Memory.

MRAM stores information using magnetic states. The resistance of a magnetic memory element represents the stored data.

One of the major advantages of MRAM is its combination of fast access, non-volatility, and high endurance. Since data remains stored even when power is removed, MRAM can reduce the need to repeatedly load data from slower storage.

RRAM

RRAM stands for Resistive Random Access Memory.

RRAM stores information by changing the resistance of a material. Different resistance states represent different data values.

RRAM is attractive because of its potential for high density, fast switching, low power consumption, and scalability. It is being investigated for applications ranging from embedded memory to emerging storage-class and neuromorphic computing systems.

Access Time

Access time is the time required to read or write information.

MRAM provides fast access because its memory state can be changed and sensed electronically through magnetic mechanisms without the need for conventional capacitor refresh.

RRAM can also provide fast switching because it changes between different resistance states. Its access performance depends on the specific RRAM technology and device implementation. Therefore, both technologies have the potential to provide **low-latency non-volatile memory**.

Throughput

Throughput refers to how much data can be processed or transferred within a given period.

MRAM can provide high throughput because of its fast read/write capabilities and suitability for random-access workloads.

RRAM can also provide high throughput and may offer advantages when large numbers of memory cells are integrated into dense arrays. Its scalability makes it attractive for future high-density memory architectures.

However, actual throughput depends on the memory architecture, controller, interface, array organization, and workload.

Energy Efficiency

Energy efficiency is particularly important in mobile devices, IoT systems, embedded computers, and data centers.

MRAM is non-volatile, so it does not require continuous power to retain stored data. This can reduce standby power consumption.

RRAM also has strong energy-efficiency potential because its resistive switching mechanism can operate with relatively low energy under suitable device conditions.

Thus, both technologies can contribute to lower-power memory architectures.

MRAM Advantages

- Very fast access
- Non-volatile operation
- High endurance
- Good energy efficiency
- Suitable for embedded and persistent-memory applications
- Can combine characteristics of conventional RAM and non-volatile storage

RRAM Advantages

- Non-volatile
- High-density potential
- Fast switching
- Low-power potential
- Scalable memory structure
- Potential applications in emerging computing architectures

Critical Comparison

MRAM is particularly attractive when the primary requirements are fast access, high endurance, and reliable non-volatile operation.

RRAM is particularly attractive when high density, scalability, and energy efficiency are important.

However, neither technology universally replaces all existing memory types. Their practical performance depends on manufacturing technology, device architecture, endurance requirements, cost, and application.

Example

In an embedded system that needs to retain configuration information even when power is removed, MRAM can provide fast non-volatile access.

In a high-density emerging memory system where large numbers of memory cells need to be integrated into a small area, RRAM can be attractive because of its potential density and scalability.

Conclusion

MRAM and RRAM are promising non-volatile memory technologies with low access latency, high performance potential, and good energy efficiency. MRAM has strong advantages in speed and endurance, while RRAM offers significant potential in density, scalability, and low-power operation. Their development can help bridge the performance gap between conventional volatile memories and persistent storage, making them important technologies for future hierarchical memory systems.